Needs for Lithium-ion Battery Runaway Containment

Application: all-electric, battery-powered passenger aircraft (commonly categorized as electric Vertical Take-Off and Landing [eVTOL], urban air mobility [UAM], or regional electric aircraft)

1. **Minimal Volumetric & Gravimetric Weight Penalty:** Every kilogram added reduces payload capacity or increases fuel burn. High volumetric energy density is vital for space-constrained avionics bays or EFB battery cases.
2. **Modularity:** Modern aircraft maintenance relies on Line Replaceable Units (LRUs). If a single cell degrades or experiences a localized fault, airlines refuse to scrap an entire $50,000+ battery pack. They require the ability to swap out sub-modules at the gate or in regional maintenance hangars.
3. **RTCA DO-160G: Battery Safety at extreme conditions** (*Environmental Conditions and Test Procedures for Airborne Equi*pment)
    a. **Thermal Cycling (DO-160G Section 4 & 5):** Validates that inter-cell thermal barriers do not crack, delaminate, or lose structural integrity when rapidly cycled between extreme cold (-55 C at altitude) and peak operating heat (+85^ C).
    b. **Multi-Axis Vibration & Structural Fatigue (DO-160G Section 8):** Tests equipment against continuous low-frequency and high-frequency airframe vibrations (e.g., engine rumble, turbulent flight load) over hundreds of hours. This ensures that thin aerogel blankets or mica plates do not pulverize into dust or shift out of position over time.
    c. **Operational Shock & Crash Safety (DO-160G Section 7):** Ensures the structural housing and inter-cell barriers maintain containment during high-G events (e.g., hard landings, sudden deceleration, or turbulence).
4. **The Flammability Requirement (FAR Part 25):** Every structural material, thermal barrier (e.g., aerogel pads, mica sheets, silicone foam), housing, and potting compound used inside an avionics bay or battery enclosure must pass strict **vertical burn tests** (e.g., 60-second or 12-second direct flame exposure) to ensure they are self-extinguishing and do not release toxic fumes or flame drips.
5. Low-cost: minimize added $/kWh and aircraft system cost.
    a. **Economic constraint:** Battery-powered flight must achieve cost parity with conventional aircraft, accounting for battery production and replacement, electricity costs, and jet-fuel costs.
6. For insurers: there is likely some threshold of guarantee, like an actual percentage, that insurers require the battery to not explode in order for them to be willing to insure it. Or, the insurance would just be insanely expensive.

The king has arrived.

Requirements for Lithium-ion Battery Runaway Containment

Application: all-electric, battery-powered passenger aircraft (commonly categorized as electric Vertical Take-Off and Landing [eVTOL], urban air mobility [UAM], or regional electric aircraft)

1. Minimal Gravimetric and Volumetric Penalty
    a. High gravimetric energy density (primary constraint): The battery pack must provide sufficient usable energy per unit mass to meet the aircraft's range and payload requirements. For a 19-seat, all-electric commuter aircraft, a representative target is approximately 650 Wh/kg at the pack level. This is substantially above current commercial battery-pack capabilities.
    b. High volumetric energy density (secondary constraint): The battery must store sufficient energy per unit volume to fit within the aircraft's available space without compromising passenger capacity, payload, or aerodynamics. A representative 19-seat commuter-aircraft design target is approximately 330 Wh/L at the pack level.

Primary source: https://doi.org/10.1038/s41586-021-04139-1
Source for b: https://doi.org/10.1073/pnas.2111164118

2. **Crashworthiness and Post-Crash Survivability:**
    a. External emergency cutoff for the battery.
    b. A lot of the safety requirements came from helicopter safety requirements, so an example is Each energy storage system, or the most critical energy storage system, should be subject to a drop-test using the following methodology:
        i. (a) the drop height should be at least 15.2 m (50 ft);
        ii. (f) after the drop test there should be no risk of post-crash fire or other harmful release within a time frame compatible with the rescue of seriously injured occupants.
            1. (1) For liquid or gaseous fuels: no leakage of flammable fluids or gases.
            2. (2) For batteries:
                a. (i) structural damage should not lead to a fire, leakage of harmful fluids, fumes or gases; or
                (ii) any fire or leakage of harmful fluids, fumes or gases should be contained for at least 15 minutes in non-occupied areas and outside the evacuation path.

3. **Low cost**
    - **Long-term target:** Battery-pack cost of approximately **$100/kWh or less**, with pack-level specific energy of approximately **800 Wh/kg** and sufficiently low electricity prices to achieve operating-cost parity on the target flight mission.

Primary source: Schäfer et al., "Technological, economic and environmental prospects of all-electric aircraft," *Nature Energy* 4, 160–166 (2019), DOI: 10.1038/s41560-018-0294-x (MIT/UCL).

4. **Consistency Requirements:** Insurers, guarantees or probabilities that things will or won't happen, and how they are demonstrated. Also to pass FAA regulations. https://www.easa.europa.eu/en/document-library/easy-access-rules/online-publications/easy-access-rules-small-category-vca?page=16#_Toc256000109 https://www.htfmarketintelligence.com/press-release/north-america-evtol-battery-failure-insurance-market
    a. Probability & Failure Rates: Failure Probability Criteria (14 CFR 25.1309 / EASA SC-VTOL):
        i. Auxiliary / APU Batteries: Loss of system safety due to single-cell runaway must be shown to be $< 10^{-7}$ per flight hour.
        ii. All-Electric / Propulsion Batteries: A catastrophic thermal event resulting in total loss of thrust or structural hull failure must meet the "Extremely Improbable" standard, defined as $< 10^{-9}$ failures per flight hour.
5. **Thermal Non-Propagation Thresholds: Single-Cell Thermal Runaway Test** (RTCA DO-311A Section 2.4.5 / FAA Special Conditions): the thermal barrier must limit heat transfer from a failing cell to surrounding equipment. The trigger cell is driven into thermal runaway (via overcharge, needle penetration, or patch heater) and typically exceeds **+800 C to +1,000 C**.
    a. **Adjacent Cell Maximum Surface Temperature:** Thermal insulation must keep adjacent cells safely below **+150 C** (or below the cell manufacturer's designated thermal runaway onset temperature, typically **+150 C to +180 C)** to prevent catastrophic chain-reaction propagation.
    b. **Observation Window:** The containment system must hold these temperature boundaries and show **zero flame propagation for a minimum of 30 minutes** post-event without active cooling or external intervention.
6. **Gas Venting, Volumetric Expansion, and Pressure Limit:** Under RTCA DO-311A and FAA SC 25-412-SC, off-gas management must prevent structural enclosure rupture and toxic cabin contamination:
    a. **Overboard Vent Exhaust Pressure:** Internal module pressure burst discs / relief valves must actuate within 0.5 to 2.0 psi (3.5 to 13.8 kPa) to route gas directly into dedicated exhaust ducts before internal pressure compromises module walls.
    b. **Gas Venting Flow Rate Volume:** Containment ducting must accommodate peak gas generation rates of **100 to 300 L/min per cell** during primary decomposition, consisting primarily of hydrogen (H_2), carbon monoxide (CO), and volatile organic solvents.

c. **Flammable Gas Dilution Limit:** Duct purging or secondary venting must maintain gas concentrations below **25% of the Lower Explosive Limit (LEL)** for hydrogen (H_2 LEL = 45 by volume in air) within any unvented airframe bays to prevent deflagration.
d. **Cabin Toxicity Thresholds:** In the event of a leak into pressurized zones, continuous exposure limits for toxic species must not exceed:
    i. **Carbon Monoxide (CO):** Max **50ppm long-term / 200 ppm** short-term ceiling.
    ii. **Hydrogen Fluoride (HF):** Max **3 ppm**

7. **Material Flammability Requirements** (14 CFR § 25.853): All inter-cell barriers, aerogel blankets, mica shields, and structural housings must pass FAR Part 25 Appendix F Bunsen burner flame testing:
    a. **Vertical Burn Test (25.853 Appendix F Part I):**
        i. Flame exposure: **60 seconds** (or 12 seconds for non-structural internal liners).
        ii. Flame temperature: **Minimum 843 C** (1,550 F)
        iii. Burn length: Average burn length $\leq$ **3 inches (76 mm)**.
        iv. After-flame time: Average self-extinguishing flame time $\leq$ **15 seconds** after removal of the burner.
        v. Drip flame time: Dwell time of flaming drips on the test floor $\leq$ **3 seconds**.
8. **Environmental & Structural Hardening Limits** (RTCA DO-160G): containment assemblies are pre-conditioned through harsh operational environmental cycling to ensure structural integrity:
    a. **Thermal Cycling (DO-160G Section 4 & 5):**
        i. Extreme operating temperatures: **-55 C +85 C** at simulated pressure altitudes up to $70,000ft$.
        ii. Ramp rates: **2 C/min to 10 C/min** across continuous multi-dwell cycles.
    b. **Vibration Profiles (DO-160G Section 8 Category R/S):**
        i. Frequency spectrum: Broadband random vibration from **5 Hz to 2,000 Hz** across 3 orthogonal axes (X, Y, Z).
        ii. Spectral density: Power Spectral Density (PSD) from **0.01 g^2/Hz to 0.1 g^2/Hz** (yielding continuous RMS acceleration of **1.5 g_{rms} to 6.0 + g_{rms} for 3 hours per axis**.
    c. **Crash Safety & Operational Shock (DO-160G Section 7):**
        i. Operational Shock: **6g peak** sawtooth pulse with **11ms** duration (3 shocks per direction across 6 directions).
        ii. Crash Safety Impulse: **20g peak acceleration** sawtooth pulse (**11 ms** duration) to guarantee that containment structures and internal inter-cell barriers maintain physical containment during emergency impact events.

Easy Access Rules for small category VCA | EASA

Application: all-electric, battery-powered passenger aircraft (commonly categorized as electric Vertical Take-Off and Landing [eVTOL], urban air mobility [UAM], or regional electric aircraft)

1. **Thermal Non-Propagation Thresholds: Single-Cell Thermal Runaway Test** (RTCA DO-311A Section 2.4.5 / FAA Special Conditions) - Thermal insulation must keep adjacent cells safely below **+150 C** (or below the cell manufacturer's designated thermal runaway onset temperature, typically **+150 C to +180 C)** at simulated pressure altitudes up to **70,000 ft** (DO-160G Section 4 & 5). There should be zero flame propagation and temperature maintenance for ≥ **30 minutes.**
2. **Post-Crash Survivability:** a 15.2 m drop and not pose a risk to human life, i.e. not leak gas, not fly into the cockpit, not become deformed and therefore a potential bomb. Needs to survive crash conditions 6g (operational shock) to 20 g (crash impulse) for 11 ms duration (DO-160G Section 7)
3. **High gravimetric energy density (primary constraint):** The battery pack must provide sufficient usable energy per unit mass to meet the aircraft's range and payload requirements. For a 19-seat, all-electric commuter aircraft, a representative target is approximately 650 Wh/kg at the pack level. Weight requirement on module weight.
4. **Flammable Gas Dilution Limit:** Duct purging or secondary venting must maintain gas concentrations below 25% of the Lower Explosive Limit (LEL) for hydrogen (H_2 LEL = 4% by volume in air) within any unvented airframe bays to prevent deflagration.
5. **Early Thermal Runaway Detection and Fail-Safe Warning Systems:** Regulatory basis: FAA AC 20-184A (Appendix C, Special Conditions 7–9; Appendices D and G), 14 CFR §§25.1309 and 25.1322, RTCA DO-311A §2.1.4.2, RTCA DO-160G, and EUROCAE ED-289. The battery monitoring system must detect abnormal cell temperature, voltage, or gas emissions and **alert the pilot within 10 seconds**, with critical sensor **readings updated at least once per second (1 Hz)** and **automatic charging disconnection** upon confirmed overtemperature or battery failure. Detection and warning systems must undergo RTCA DO-160G environmental testing and fault-injection testing, demonstrating reliable operation under temperature extremes, vibration, altitude changes, and individual sensor failures, with zero missed required warnings across the qualification test matrix.

Flame propagation prevention in batteries technologies

Possible: combination of venting and materials that change conductivity at different temps, then crash that bih. venting has been shown to be great at regulating module temperature, and this thermally responsive material would allow for ONE module undergoing thermal runaway to suddenly become insulated while the venting mechanism still cools other modules.
We can also explore putting sensors that detect thermal runaway INTO the venting channels, but that might be OD for an 8 week lil research project.

- Ceramic coatings?
- Phase change materials
- Microlayers or super thin coatings
- Aerogelllllll - Knudsen effect, pore size (20 - 40 mm) smaller than the mean free path
- Making sure vents don't get clogged with loose materials or particles
- Prevention of gas propagation or making sure gas that escapes a battery is properly handled
- Passive localized cooling, so it somehow reroutes cooling to hot modules
- Active discharge of thermal runaway module without getting rid of the baseline energy levels required to maintain propulsion
- Taking coolant from other parts of the aircraft in an emergency to quickly prevent thermal runaway without the other parts of the the aircraft getting slimed out
- Discover some new perfect material that has EVEN BETTER thermal reactive conduction responsive thingy.

Gap: venting dynamics and flame quenching needs to be evaluated at flight altitudes and pressure drops. Low pressure increases gas volume expansion, velocities, and mass flow rates. Need to build strong vents to withstand.
Gap: Low atm reduces physical barrier convection cooling efficiency
Gap: Existing papers do not offer ultra-lightweight < 5 - 8%) integrated duct-barrier solutions that meet FAA 10^{-9} safety probability metrics without compromising flight range.

Beta Battery - https://beta.team/battery
https://www.nature.com/articles/s41560-024-01535-5 - temperature responsive material 2024
https://ideas.repec.org/a/eee/rensus/v239y2026ics136403212600448x.html - review on thermal runaway propagation prevention in aircraft batteries
https://iopscience.iop.org/article/10.1149/1945-7111/adc50f - venting batteries APL 2025

Multiple materials (conductive, insulating, phase change) and reaction heat and pressure-triggered venting.
https://www.sciencedirect.com/science/article/pii/S259011682600086X
Battery thermal stability - https://xray.greyb.com/ev-battery/thermal-stability-nmc

- This website is an exhaustive list of a bunch of battery properties

Thermal isolation

Lava rock -
https://heatshieldproducts.com/product/automotive/component-heat-shields/battery-heat-shield/

Thin aerogel -https://aerogelzone.com/thin-aerogel-thermal-barrier/
Aero gel for prototyping - https://aerogel-it.shop/collections/home
Mica insulators - https://www.ashevillemica.com/automotive-mica/

Thermal Insulation and Fireproof Aerogel Composites for Automotive Batteries - https://www.mdpi.com/2310-2861/11/10/791

Organic and Inorganic Hybrid Composite Phase Change Material for Inhibiting the Thermal Runaway of Lithium-Ion Batteries https://www.mdpi.com/2313-0105/9/10/513

1. Electrical isolation: stopping "current dumping" between parallel cells

Your list only addresses heat. In parallel-connected packs, a failing cell can also be fed by its neighbors' current. One study found that packs with electrically connected cells propagated thermal runaway because neighboring cells dumped current into the failing one, while packs with electrically disconnected cells did not propagate and neighbors stayed below 85 °C. That work led to the first battery design with individual cell fusing built into the current collectors. The approach adds very little mass. A related idea is thermally disconnecting busbars, since copper busbars between tightly packed modules can become a heat transfer path during a fire.
https://ideas.repec.org/a/eee/appene/v286y2021ics0306261921000568.html IDEAS home Printed from https://ideas.repec.org/a/eee/appene/v286y2021ics0306261921000568.html +2

2. Endothermic "sacrificial" chemistry (different from PCMs)

Instead of melting, these materials decompose chemically and absorb heat at runaway temperatures. Aluminum hydroxide, for example, releases its crystal water when it decomposes, absorbing heat, and leaves behind porous alumina that then acts as an insulator. Decomposition onset can be matched to the runaway window: about 200 °C for aluminum hydroxide and about 330 °C for magnesium hydroxide. This gives you a heat sink and an insulator in one material,

which could be a low-mass complement to your thermally responsive material idea. GM has a patent on this approach.
https://patents.justia.com/patent/20240106021 usptouspto

3. Intumescent materials

These are distinct from ceramic coatings because they swell into an insulating char only when heated, so they are thin and light during normal operation. In an early Tesla patent, uncoated cells propagated runaway to the next cell within 3 to 4 minutes, while intumescent-coated cells did not propagate. A Warwick study comparing five barriers found a stainless steel barrier coated with intumescent flame-retardant fabric performed best, and sidewall rupture counts correlated linearly with propagation rate. That makes it relevant if you're using high-energy cylindrical cells.
https://wrap.warwick.ac.uk/186891 usptowarwick

4. Flame arrestors and quenching meshes in the vent path

This addresses your altitude and venting gap directly. A conductive mesh over the vent cools exiting gas below its auto-ignition temperature, and the required hole size depends on the gas's auto-ignition temperature and velocity. There is a trade-off you'll care about at low pressure: smaller openings reject more heat but increase the forces on the mesh during a vent. A 2025 BIT study designed a multilayer quenching plate and found that front-facing configurations needed five layers at 75% SOC and seven at 100% SOC, while side-facing configurations needed only three. Re-evaluating that layer count at altitude-level gas velocities could be a well-scoped 8-week question.

https://pure.bit.edu.cn/en/publications/design-of-a-multilayer-flame-arrestor-for-lithium-ion-batteries-a/ Low profile pressure disconnect device for lithium ion batteries +2

5. Self-triggered suppression microcapsules

This is a passive alternative to your "borrow coolant from the aircraft" idea, with no plumbing. The capsules rupture at 130 °C and release Novec 1230, which extinguishes flames, and HFO-1336mzz(Z), which provides cooling; together they delayed runaway and inhibited propagation. Earlier UMD work found that Novec 1230 at 15.2 vol.% significantly inhibited combustion of ejected material and prevented complete propagation through fully charged cell arrays.
https://pubs.rsc.org/en/content/articlehtml/2025/ta/d5ta03776j rscumd

6. Oxygen exclusion and inerting

This approach removes oxygen from the fire triangle instead of fighting heat. In one cited test, propagation occurred at normal 21% oxygen, but no significant propagation to adjacent cells occurred when oxygen was reduced to about 11.3%. The caveat is that nitrogen doesn't instantly penetrate sealed cells, so inerting suppresses the fire around the cell rather than the runaway inside it. A sealed module with a one-way vent, which keeps oxygen out while letting gas escape, is a lighter variant of the same idea.
https://www.aidic.it/cisap12fex26/programma/22Renzi.docx aidicaidic

7. Two-phase dielectric immersion

Cells sit in a fluid that boils at a chosen temperature. In Carrar's test, a 72 Ah pouch cell was driven above 800 °C while the adjacent cell stayed at about 50 °C, and the company says the system works passively, without sensors or suppression systems. The fluid mass is the obvious problem for your weight budget, but it is a useful benchmark to compare against.

https://chargedevs.com/newswire/carrar-says-its-immersion-cooled-battery-architecture-stops-thermal-runaway-even-with-a-cell-above-800-c/ chargedevschargedevs

8. Heat pipes that vent and become insulators

This is closest to your thermal-switch concept. In this patent, a heat pipe cools cells normally. After a thermal event, a pressure-relief device lets it dump its working fluid to become a vented heat pipe, which then inhibits heat transfer to the next cell. So it conducts heat when things are normal and insulates during a failure, using existing hardware.
https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/9689624 uspto

9. Intra-cell safety: stopping the first cell from running away

Stanford developed a current-collector coating that shuts the battery down under overheating or shorting and lets it resume normal function afterward, with a switching temperature that can be tuned. On the electrolyte side, a nonflammable localized high-concentration electrolyte raised the runaway trigger temperature by 47.3 °C and lowered the peak runaway temperature by 71.8 °C. A caution: a Nature Communications study found that non-flammable concentrated electrolytes could not prevent runaway, so "nonflammable" does not mean "safe."
https://ideas.repec.org/a/nat/natene/v1y2016i1d10.1038_nenergy.2015.9.html IDEAS home Printed from https://ideas.repec.org/a/nat/natene/v1y2016i1d10.1038_nenergy.2015.9.html +2

10. Pack geometry and directional venting

This is free in terms of mass. A CFD study of an LFP module found that inverting the cells and exhausting from both the side and bottom of the module was the best venting scheme. TUM

work also showed that vented particles, not just gas, significantly affect propagation in enclosed packs, which ties into your vent-clogging concern.
https://doaj.org/article/7c2a909c5188428a93eb51ca06bb1ed9 doajtum

11. Off-gas detection before runaway

This is a simpler, off-the-shelf version of your "sensors in vent channels" idea. In third-party testing, an off-gas sensor gave 6.4 minutes of warning before runaway, and stopping the charge when it alarmed avoided runaway entirely. Notably, a hydrogen sensor only detected gas at the moment of runaway. This pairs naturally with your "active discharge" idea, because the warning gives the system time to act.

https://www.honeywell.com/us/en/insights/articles/national-lab-testing-demonstrates-value-in-early-detection

Materials

- Ceramics
- Phase change materials
    - Organic and Inorganic Hybrid Composite Phase Change Material for Inhibiting the Thermal Runaway of Lithium-Ion Batteries https://www.mdpi.com/2313-0105/9/10/513
- Aerogel
- https://www.nature.com/articles/s41560-024-01535-5 - temperature responsive material 2024
- Mica insulators
- **Endothermic "sacrificial" chemistry - Aluminum hydroxide**
    - https://patents.justia.com/patent/20240106021 usptouspto
- **Intumescent materials - painting/coating**
    - https://wrap.warwick.ac.uk/186891 usptowarwick
- Fast and reversible thermoresponsive polymer switching materials for safer batteries
    - https://ideas.repec.org/a/nat/natene/v1y2016i1d10.1038_nenergy.2015.9.html IDEAS home Printed from https://ideas.repec.org/a/nat/natene/v1y2016i1d10.1038_nenergy.2015.9.html +2
- Lava rock - https://heatshieldproducts.com/product/automotive/component-heat-shields/battery-heat-shield/

Tested Material Combinations in Literature

Multiple materials (conductive, insulating, phase change) and reaction heat and pressure-triggered venting. https://www.sciencedirect.com/science/article/pii/S259011682600086X

Research into mitigating lithium-ion battery thermal runaway propagation has increasingly focused on multi-functional hybrid systems that pair static thermal insulation with active enthalpy absorption or dynamic thermal regulation.

- **Aerogel + Phase Change Material (PCM) Composites**
    - *Mechanism*: Combining nanoporous silica or nanofiber aerogel matrices impregnated with organic/inorganic phase change materials (CPCMs). The PCM acts as an active latent heat sink during early-stage thermal spikes, while the inorganic aerogel framework prevents liquid leakage and provides permanent low thermal conductivity (approx 0.013 - 0.018 W/m*K) once phase transition occurs.
    - *Literature Reference*: Mei et al. (2023) developed organic-inorganic hybrid CPCM aerogel matrices, demonstrating effective suppression of thermal propagation. Similarly, Wong et al. (2024) evaluated nanofiber aerogel composite materials integrated with phase-absorption layers to delay time intervals between runaway events in high-energy modules.
- **Temperature-Responsive Polymer Switch + Insulation Interlayers**

    - *Mechanism*: Advanced thermal-switching materials that dynamically alter their thermal conductivity (transitioning from a conductive state during normal operation to an adiabatic insulating state during high-temperature anomalies).
    - *Literature Reference*: Wang et al. (2024) designed a rapid temperature-responsive thermal regulator that prevents up to 80% of heat transmission during thermal runaway while maintaining uniform temperature distribution under normal working conditions.

2. Phase-Change Material (PCM) + Intumescent Polymer (Enthalpy Buffer + Expandable Shield)

- **How They Work Together:** Encapsulate cells in a composite PCM wax/salt matrix, and coat module divider walls with an intumescent polymer coating.
- **Synergy:** The PCM acts as an active heat sink during early-stage thermal spikes, absorbing latent thermal energy at a constant temperature before thermal runaway fully triggers. If the trigger cell exceeds the PCM's heat capacity and ignites, the extreme heat causes the intumescent coating to expand up to 100×, creating a secondary char barrier that blocks runaway flame propagation to the next cell tier.
- **Primary Application:** Dense pouch or cylindrical cell arrays. explored in advanced thermal protection systems, particularly for dense battery packs and aerospace applications where space is tightly constrained.
- **Weaknesses:** Significant volumetric expansion requires mechanical spacing allowances within the module design. Liquid PCM leakage risks and chemical incompatibility between organic waxes and polymer binder matrices can also complicate manufacturing.
- **Unexplored Areas:** Long-term thermal-cycling durability under repeated charge/discharge loads (where repeated melting and solidifying of the PCM can degrade surrounding intumescent bonding integrity) remains underexplored in literature.

3. Ceramic Fiber Paper + Steel Structural Framing (Thermal Separation + Mechanical Containment)

- **How They Work Together:** Line steel compartment walls or structural module frames with compressible aluminosilicate ceramic fiber paper.
- **Synergy:** Steel provides high structural rigidity, containment strength, and blast resistance, but its high thermal conductivity causes heat to rapidly transfer to neighboring compartments. Thin ceramic fiber paper insulates the steel surfaces, preventing thermal bridging while retaining the structural enclosure benefits.
- **Primary Application:** Sub-module bulkheads and module housing frames.
- **Weaknesses:** Ceramic fiber papers are friable, prone to shedding particulates under vibrational stress, and add non-active weight that degrades the gravimetric energy density of the battery pack.
- **Unexplored Areas:** Optimization of ultra-thin, binder-modified ceramic papers that maintain high compression resilience under high-vibration aerospace or EV operating environments while completely eliminating particulate dusting.

4. Intumescent Aerogel Composites (Integrated Dual Barrier)

- **How They Work Together:** A single manufactured composite material featuring a nanoporous aerogel core impregnated or laminated with intumescent expanding compounds.
- **Synergy:** Combines the low static thermal conductivity of aerogel with dynamic thermal insulation—as temperatures exceed 200°C, the intumescent component expands to fill air gaps left by cell swelling or geometric deformation during runaway.
- **Weaknesses:** High production costs, multi-step chemical synthesis complexity, and potential brittleness of the aerogel framework under mechanical shear stress.
- **Unexplored Areas:** Scalable, low-cost impregnation techniques for embedding intumescent nanoparticles directly into flexible aerogel blankets without destroying the delicate 3D nanoporous pore structure.

Untested or Novel Material Combinations (Ranked from Most to Least Promising)

Below are innovative combinations of the listed materials that have not been heavily explored or published in standard configurations, ranked by overall potential impact.

1. Silica Aerogel + Endothermic Aluminum Hydroxide + Phlogopite Mica

- **Concept:** Sandwiching an ultra-thin silica aerogel layer infused with aluminum hydroxide ($Al(OH)_3$) powder, backed by a rigid phlogopite mica flame shield.
- **Strengths:**
  - Provides a triple-defense mechanism: aerogel stops conductive heat transfer, aluminum hydroxide undergoes an endothermic decomposition releasing cooling water vapor ($Al(OH)_3 \rightarrow Al_2O_3 + 3H_2O$) upon reaching high temperatures, and mica deflects direct high-velocity plasma/flame jets.
- **Weaknesses:**
  - Increased thickness and weight penalty; potential manufacturing challenges in uniformly distributing aluminum hydroxide within brittle aerogel structures.

2. Phase Change Material (PCM) + Intumescent Coating + Basalt ("Lava Rock") Fiber Wrap

- **Concept:** Encapsulating cells in a PCM matrix while wrapping the module assembly in a basalt fiber fabric coated with an intumescent paint.
- **Strengths:**
  - Basalt fiber offers superior radiant heat resistance and structural integrity under fire conditions. If the PCM saturates and a cell breaches, the intumescent coating expands up to $100\ \times$ to seal air gaps and trap expanding gases.
- **Weaknesses:**
  - Significant volumetric expansion requires dedicated mechanical spacing tolerances; heavy multi-layer setup reduces pack gravimetric energy density.

3. Aluminosilicate Ceramic Fiber Paper + Aluminum Hydroxide Slurry Impregnation

- **Concept:** Impregnating porous high-temperature ceramic fiber paper with an endothermic aluminum hydroxide binder solution.

- **Strengths:**
    - Ceramic fiber maintains structural stability above $1,200°C$, while the sacrificial endothermic chemistry absorbs massive thermal energy during decomposition.
- **Weaknesses:**
    - Ceramic fiber papers are friable and prone to shedding particulate matter; water vapor release from aluminum hydroxide during accidental mild overheating could potentially interact with compromised cell packaging.

4. Basalt ("Lava Rock") Fiber Fabric + Mica Plate Lamination

- **Concept:** Mechanically stitching or bonding basalt fiber textile layers directly to rigid phlogopite mica plates.
- **Strengths:**
    - High flexibility, ease of cutting/prototyping, and exceptional resistance to open flames and electrical arcing.
- **Weaknesses:**
    - Lacks intrinsic low thermal conductivity (both basalt and mica conduct heat faster than aerogels) and provides zero active latent heat buffering.

Carbon Fiber Reinforced Polymer (CFRP) & Lightweight Composites

Carbon Fiber Reinforced Polymer (CFRP) and lightweight composite materials are heavily utilized in aerospace engineering due to their exceptional strength-to-weight ratios. When applied to lithium-ion battery containment and thermal runaway prevention, they serve as lightweight structural housings or localized cell sleeves.

Have Been Done

1. **CFRP Structural Enclosures + Internal Aerogel / Mica Blankets**
    - **How They Work Together:** The aircraft battery module uses a lightweight molded CFRP outer box or tray for primary structural rigidity, lined internally with thin nanoporous silica aerogel sheets and phlogopite mica plates.
    - **Strengths:** CFRP provides high specific strength and stiffness without the heavy gravimetric penalty of steel or thick aluminum, while the internal aerogel and mica layers provide thermal insulation and flame deflection.
    - **Weaknesses:** CFRP matrices are organic polymer-based and can degrade, delaminate, or burn when subjected to prolonged direct high-temperature flame jets (>600 C) unless protected by an inner thermal barrier.
2. **NASA-Style CFRP Cell Sleeves / Wraps**
    - **How They Work Together:** Individual cylindrical cells are wrapped in a thin, custom-engineered carbon fiber sleeve infused with flame-retardant epoxy matrices.
    - **Strengths:** Prevents sidewall ruptures (SWR) of small cylindrical cells, adds minimal mass compared to metal housings, and is non-electrically conductive depending on the weave and matrix sizing.

Not Yet Been Done (Unexplored or Emerging Areas)

1. **Intumescent-Impregnated CFRP Matrix Composites**
    - **Concept:** Embedding expandable intumescent microcapsules directly into the epoxy resin matrix of the structural CFRP weave during manufacturing.
    - **Strengths:** During normal flight, the CFRP functions purely as a high-strength structural skin. If a localized thermal runaway spike occurs, the embedded intumescent compounds activate, swelling internally to choke off oxygen and create a thick carbonaceous char barrier within the composite plies themselves.
    - **Weaknesses:** Introducing foreign compounds into the epoxy matrix can compromise the interlaminar shear strength and mechanical fatigue life of the composite under cyclic aircraft loading.
2. **CFRP + Aerogel-Core Sandwich Panels with Integrated Micro-Venting Channels**
    - **Concept:** A structural sandwich panel consisting of thin CFRP face sheets enclosing a core made of intumescent aerogel composite, integrated with micro-machined venting flutes that safely channel off-gases overboard.
    - **Strengths:** Offers a multi-functional structural wall that simultaneously handles mechanical crash loads, provides ultra-low thermal insulation, and acts as a pressure-relief duct manifold for aviation environments.
    - **Weaknesses:** Extremely complex and costly multi-step manufacturing process; difficult to nondestructively inspect for internal micro-delamination after vibrational stress.

Key Trade-Offs for Aviation (eVTOL / Commuter Aircraft)

- **Thermal-Structural Conflict:** While CFRP provides exceptional weight savings to help reach pack targets ($\approx 650\ Wh/kg$), standard polymer matrices lose structural integrity at elevated temperatures (>300- 400 C).
- **Regulatory Compliance:** Any CFRP housing or internal composite barrier used in passenger aircraft must satisfy strict flammability, smoke, and toxicity (FST) mandates such as FAR 25.853 and UL 94 V-0, requiring specialized flame-retardant resin systems (e.g., modified cyanate esters or high-temperature epoxy syntactics).

References

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Three-Material Hybrid Barrier Combinations for Electric Aviation

1. **Carbon Fiber Reinforced Polymer (CFRP) Plate** + Nanoporous Silica Aerogel Sheet + Phlogopite Mica Plate
    a. Using a thin CFRP plate (high stiffness) as the structural outer wall of the barrier box, lined internally with aerogel (highly flexible) and mica (stiff plate, note: can crack/fracture under bending stress).
2. **PCM Composite Matrix** + Intumescent Polymer Coating + Compressible Ceramic Fiber Paper
    a. Packing the space between the steel rod and Target Cell 2 with a PCM (semi-rigid to waxy, minimal structural stability) matrix and lining the walls with intumescent-coated ceramic paper.
    b. Intumescent Polymer Coating (rubbery, low structural rigidity), Compressible Ceramic Fiber Paper (soft, blanket-like cushion)
    c. **Setup Limitation:** Option B uses a controlled electric cartridge heater, meaning there is no real chemical off-gassing or raw flame to trigger the intumescent expansion properly. An intumescent coating relies on extreme flash heat or direct flame to expand 100 ×; a clean electric steel rod might not trigger full intumescent expansion unless the setpoint exceeds 300 –400 C, making this combination harder to evaluate cleanly without a real thermal runaway trigger.
3. Ceramic Fiber Felt + Silica Aerogel + Phlogopite Mica (Flexible Inorganic Sandwich Barrier)
    a. No outer structural frame
4. Combination: Aerogel + Phlogopite Mica (Flexible + Rigid Hybrid)
    a. An ultra-thin flexible aerogel blanket is placed against the steel rod/cell face to conform to the geometry, backed by a rigid phlogopite mica plate clamped inside the barrier wall slot.

Combination 1: Structural-Thermal-Flame Barrier

Materials: *CFRP Plate + Nanoporous Silica Aerogel Sheet + Phlogopite Mica Plate*

Material-by-Material Reasoning

1. **Carbon Fiber Reinforced Polymer (CFRP) Plate**
    - *Reasoning:* Serves as the primary structural housing and load-bearing inter-module wall. Unlike heavy steel or aluminum, CFRP provides an exceptional specific strength-to-weight ratio, ensuring the battery pack satisfies aviation gravimetric energy density goals without compromising structural rigidity.

2. **Nanoporous Silica Aerogel Sheet**
    - *Reasoning:* Implemented as the core static thermal insulator. With extreme porosity (≈99%) and an ultra-low thermal conductivity (k≈0.013–0.018 W/m·K), it blocks conductive heat flux to keep adjacent cell skin temperatures safely below the 150◦C non-propagation threshold during a thermal event.
3. **Phlogopite Mica Plate**
    - *Reasoning:* Acts as a sacrificial high-temperature (>1,000◦C) flame and plasma shield. Raw aerogels can erode under direct high-velocity particle ejecta; the mica plate deflects incandescent jets and provides essential dielectric insulation against electrical arcing.

- **Strengths:**
    - Exceptional weight-to-strength performance tailored for aerospace structural demands.
    - Robust multi-stage defense against both slow conductive heat transfer and high-velocity plasma jets.
    - **Cell Deformation Behavior:** The rigid exterior Carbon Fiber Reinforced Polymer (CFRP) housing combined with internal mica plates provides a strong mechanical box structure. This constrains normal cell swelling and prevents neighboring cells from bulging into one another during high-rate aviation discharge cycles.
    - **Post-Crash Survivability Probability: Very High:** CFRP offers an exceptional strength-to-weight ratio that absorbs high-G crash impulses without yielding or adding excessive dead weight (preserving the $\approx 650 Wh/kg$ pack target). The outer composite housing prevents the battery array from breaking apart or deforming into a "bomb" structure during a crash event, satisfying DO-160G crash safety requirements while the inner aerogel and mica layers maintain thermal stability.
- **Weaknesses:**
    - CFRP polymer matrices degrade if exposed directly to prolonged temperatures above 200◦C–300◦C without protective facings.
    - Rigid mica and CFRP plates require precise mechanical tolerances to accommodate normal cell swelling. Because the CFRP and mica layers are rigid, they lack high elasticity. If a cell undergoes severe swelling or mechanical deformation during early-stage abuse, rigid plates can create localized pressure pinch-points, potentially accelerating internal short circuits.

Resembling Literature & Work: Research into lightweight structural composite enclosures paired with multi-layer thermal barriers. Studies evaluate how combining structural composites (like carbon fiber or high-strength polymers) with internal low-conductivity aerogel pads and mica flame shields prevents thermal propagation while maintaining structural integrity.

- *Liu et al. (2022 / 2024) – "Experimental study on the alleviation of thermal runaway propagation from an overcharged lithium-ion battery module using different thermal insulation layers"* (*Energy* / *Journal of Energy Storage*), which examines the synergy of modular thermal insulation layers and high-temperature mineral shields.
- *PatSnap Eureka EV Safety Reports (2026)* – Detailing structural composite enclosures and passive thermal barriers utilizing aerogel felts and mica compression plates for electric vehicle and aviation modules.

Combination 2: Enthalpy-Dynamic-Cushion Barrier

Materials: *PCM Composite Matrix + Intumescent Polymer Coating + Compressible Ceramic Fiber Paper*

Material-by-Material Reasoning

1. **Phase-Change Material (PCM) Composite Matrix**
    - *Reasoning:* Functions as an active latent heat sink. During heavy aircraft discharge cycles (e.g., 3C–5C takeoff/climb) or early thermal spikes, the PCM absorbs massive amounts of thermal energy nearly isothermally via phase transition, smoothing out operating temperatures.
2. **Intumescent Polymer Coating**
    - *Reasoning:* Provides dynamic thermal expansion. When subjected to temperatures exceeding critical thresholds (≈200◦C), the intumescent binder swells up to 100× its original volume, generating a thick carbonaceous char layer that autonomously seals air gaps created by cell deformation.
3. **Compressible Aluminosilicate Ceramic Fiber Paper**
    - *Reasoning:* Acts as a high-temperature structural cushion and thermal break. It withstands continuous temperatures above 1,000◦C while absorbing multi-axis aircraft vibrations (DO-160G Section 8) and buffering mechanical expansion cycles.

- **Strengths:**
    - Dynamic gap-filling and active enthalpy buffering prevent thermal spikes from escalating into full runaway.
    - Compressible ceramic layers absorb mechanical shock and vibration effectively under flight conditions.
    - Cell Deformation Behavior: Highly compliant. The compressible ceramic fiber paper and soft PCM/intumescent matrix act as mechanical buffers. They naturally absorb and accommodate the breathing, expansion, and minor geometrical deformations of pouch or prismatic cells over their operational lifetime.
- **Weaknesses:**

    - PCMs add non-active weight and present potential liquid leakage risks if encapsulation fails under extreme mechanical shock.
    - Intumescent expansion requires dedicated void space within the module architecture to prevent over-pressurization.
    - Under massive structural impacts or crashes, soft or semi-fluid materials provide virtually zero structural rigidity on their own, leaving the cells vulnerable to crushing unless housed in a rigid secondary frame.
- **Post-Crash Survivability Probability: Moderate.**
    - *Reasoning:* While this combination excels at internal thermal management and local gap-sealing via intumescent expansion ($100\ \times$), it lacks structural load-bearing capacity. Under a severe crash impulse ($20g$), a module relying solely on soft PCMs and ceramic paper risks bulk structural collapse, potentially causing widespread mechanical deformation across the cell array unless backed by an external aerospace-grade frame.

Resembling Literature & Work: Work focusing on active latent heat buffers combined with expandable intumescent coatings and high-temperature inorganic fiber pads to manage both normal operating heat and extreme thermal events.

Key References:

- *MDPI Reviews on Phase Change Materials for Battery Thermal Management (2026)* – Systematically evaluating composite PCM degradation, latent heat storage, and integration with fire-retardant intumescent systems.
- *US Patent US20170214103A1 – "Lithium Ion Battery With Thermal Runaway Protection"* – Describing endothermic ceramic matrices and multi-component insulation configurations that combine energy absorption with high-temperature barrier protection.

Combination 3: Ceramic Fiber Felt + Silica Aerogel + Phlogopite Mica (Flexible Inorganic Sandwich Barrier)

Material-by-Material Reasoning

1. **Ceramic Fiber Felt (CFF)**
    - *Reasoning:* Serves as a high-temperature compressible fibrous backbone (>1,000 C). It absorbs multi-axis mechanical vibrations (DO-160G) and maintains structural thickness under compression.
2. **Silica Aerogel Blanket (FA)**
    - *Reasoning:* Impregnated or layered directly within the felt matrix to provide world-class static thermal insulation (approx 0.013 - 0.018 W/m K), blocking conductive heat transfer between cells.

3. **Phlogopite Mica Roll (MR)**
    - *Reasoning:* Applied as an outer protective face sheet. It provides an impermeable, high-temperature dielectric shield that deflects direct high-velocity plasma jets and incandescent particle ejecta without eroding.

- **Strengths:**
    - 100% inorganic composition ensures zero toxic smoke emission or organic binder degradation during a fire event.
    - Exceptional flexibility and vibration resistance tailored for dynamic aerospace environments.
    - Cell Deformation Behavior: The compressible ceramic felt core functions like an engineered cushion. It dynamically absorbs normal cell swelling and multi-axis vibrational stresses (satisfying DO-160G Section 8 vibration criteria) without transmitting rigid pressure spikes back to the cell casing.
- **Weaknesses:**
    - Lacks active latent heat buffering (no PCM) during early-stage temperature spikes.
    - Slightly thicker profile than bare aerogel films due to the underlying felt and mica layers.
    - Entirely flexible systems offer no structural containment against high-pressure gas venting or violent cell expansion, meaning individual cells can still push outward if internal pressures spike.
- **Post-Crash Survivability Probability: Moderate-Low (As a standalone housing).**
    - *Reasoning:* While completely inorganic and non-flammable under extreme flame jets, a soft sandwich barrier cannot provide structural crash containment. To achieve high survivability probability in an aircraft, this flexible blanket must be paired with an outer lightweight structural frame (such as a thin CFRP or titanium exoskeleton) to prevent catastrophic pack deformation during a crash landing.

Supporting Literature References

- *PMC Research (2025) – "Thermal Insulation Efficiency and Mechanisms of Flexible Sandwich Thermal Barriers for Lithium-Ion Batteries"* (PMC11989640) – Specifically investigates flexible sandwich structures combining mica rolls, ceramic fiber felts, and fiberglass aerogels, analyzing coupled heat transfer mechanisms and cold-surface temperature rise.

Material Properties

Material	Ability to Absorb / Manage Heat	Structural Stability	Unique Traits
Carbon Fiber Reinforced Polymer (CFRP) $0.5mm$ **to** $2.0mm$ **(commonly** $1.0mm$**)**	**Low-Moderate:** Acts primarily as a structural barrier rather than a thermal sink; susceptible to epoxy matrix degradation above **200°C–300°C**.	**Very High (Rigid):** Exceptional strength-to-weigh t ratio; high in-plane stiffness and tensile strength.	Extremely lightweight; tailored for aerospace gravimetric targets ($\approx 650Wh/kg$); requires flame-retardant sizing to pass FST mandates.
Phlogopite / Muscovite Mica $0.2mm$ **to** $1.5mm$ **(commonly** $0.5mm$**)**	**Moderate-High (Thermal Shielding):** Withstands extreme temperatures (**>1,000°C**) without melting; reflects radiant heat and blocks plasma jets.	**High (Rigid Plates) / Low (Flexible Rolls):** Rigid plates maintain flat structural form; flexible rolls conform to curves but lack load-bearing strength.	Excellent natural electrical insulator (dielectric); sacrificial mineral structure that chars/deflects without emitting toxic smoke.
Nanoporous Silica Aerogel $0.5mm$ **to** $3.0mm$ **(commonly** $1.0mm$ **or** $3.0mm$**)**	**Exceptional (Thermal Barrier):** World-class static thermal conductivity; blocks conductive heat flux entirely.	**Very Low (Soft & Fragile):** Extremely porous ~99%; has near-zero structural rigidity and can powder or compress permanently under heavy loads.	The lightest solid material known; requires encapsulation or a backing layer (like mica or felt) to protect against physical erosion.

Phase-Change Materials (PCMs)	**Very High (Latent Heat Buffer):** Absorbs massive thermal energy isothermally via phase transition (melting paraffin/salts) during early thermal spikes.	**Low (Semi-Rigid to Liquid):** Waxy solid when cool, liquid when heated; provides zero structural load-bearing capacity.	Prevents transient temperature spikes from triggering full runaway; risk of liquid leakage if encapsulation fails under shock.
Intumescent Polymers / Coatings	**High (Dynamic Endothermic Char):** Absorbs heat during initial decomposition and forms an insulating carbonaceous char barrier.	**Low (Rubbery / Soft):** Functions as a coating or paint; mechanical rigidity is negligible prior to expansion.	Expands exponentially up to **100×** its original volume when exposed to heat ($\approx \mathbf{200^\circ\text{C}}$) to autonomously seal air gaps and block flames.
Ceramic Fiber Paper / Felt (Aluminosilicate) $0.5mm$ **to** $3.0mm$ **(commonly** $1.0mm$ **or** $2.0mm$**)**	**High (High-Temp Insulation):** Withstands continuous temperatures **>1,000°C** while providing a stable thermal break.	**Moderate (Compressible Cushion):** Soft and spring-like; engineered to absorb vibrations and mechanical "breathing" without rigidity.	Excellent vibration damping (DO-160G Section 8); prone to particulate dusting if unsealed.
Aluminum Hydroxide ($Al(OH)_3$**)**	**Very High (Sacrificial Endothermic):** Undergoes endothermic decomposition ($2Al(OH)_3 \rightarrow Al_2O_3 + 3H$) past **200°C**, absorbing massive heat and releasing cooling water vapor.	**None (Powder / Filler):** Must be bound within a polymer, textile, or slurry matrix to be utilized.	Acts as a chemical cooling agent right at the source of thermal abuse.

Basalt ("Lava Rock") Fiber $1.0mm$ **to** $4.0mm$ **(commonly** $2.0mm$**)**	**Moderate (Radiant Shield):** Offers good high-temperature resistance and flame barrier properties up to **~700°C–800°C**.	**Moderate (Textile Weave):** Flexible fabric that retains tensile strength under fire conditions better than standard glass fiber.	Volcanic rock-derived mineral; highly flame-resistant, eco-friendly, and cost-effective compared to specialized carbon or aramid fibers.

Layer 1 (Cell-Facing Shield): $0.5mm$ **Phlogopite Mica Sheet** (deflects direct flame jets and plasma arcing).

Layer 2 (Core Thermal Insulation): $2.0mm$ **Silica Aerogel Blanket** (blocks conductive heat transfer to keep adjacent cells below 150 C.

Layer 3 (Mechanical Cushion / Outer Wrap): $1.0mm$ to $1.5mm$ **Basalt Fiber Felt or Ceramic Paper** (absorbs cell "breathing" and multi-axis aircraft vibrations).

Total Stack Thickness: $3.5mm$ **to** $4.0mm$, leaving a safe buffer for modular assembly tolerances.

10/12 - 10/16

Currently $330

Battery Dimensions: 65.00mm; diameter max 18.40mm (2.55906 in , 0.7244094 in)

Budget: $650
ON SOME REAL SHIT: Batteries
https://www.18650batterystore.com/products/samsung-25r-18650 x 4 = 30

Cartridge heater: $20
https://oemheaters.com/product/5260/34-inch-x-25-inch-60w-120v-cartridge-heater?utm_source=chatgpt.com

Sand: $8
https://www.walmart.com/ip/Stoney-River-White-Aquatic-Sand-5-Pound-Bag/193179912?utm_medium=feed&utm_source=chatgpt.com&oppcref=1cddd204-1889-45aa-989f-f9ba63fb05ec

New rigid shmittle 0.032 inch (0.8128 mm)
https://www.mcmaster.com/85165K26/

- $77.97 each

SUPERIOR aerogel 65 bucks + 10 dollar shipping 4 day shipping, 2mm (1300 C)
https://www.buyaerogel.com/product/skogar-ht/?attribute_size=40+cm+x+30+cm+x+2+mm

- 40 cm x 30 cm
- 15.748'' x 12''

1.5 mm: https://www.buyaerogel.com/product/airloy-hr-rollable-sheet/ (300 C)

- $90 for 6'' x 1 ft

Basalt Fabric: https://www.mcmaster.com/products/~/material~basalt-fabric/?s=basalt+fiber

- Get the 2'' width
- **1.19062 mm thick,** 3/64 inch thick
- **$20.25 each for 15 ft** *OR $33.75 for 25 ft*

High-Strength Lightweight Carbon Fiber Sheet: 12" x 24" x 1/32" Part number 8181K32 $100.31
https://www.mcmaster.com/products/sheet-stock/carbon-fiber-1~/?s=carbon+fiber+sheet+stock

- Thickness: 0.79375 mm

4mm / 10/64inch total thickness

Layer possibilities:
Mica aerogel mica
Carbon fiber aerogel carbon fiber
Mica ceramic aerogel ceramic mica
Basalt aerogel basalt
Basalt mica aerogel mica basalt

Fuelcell store
McMaster?

Both blocks are 12'' inches long – Need to cut to 2''

1. Block of steel (runaway cell) - https://speedymetals.com/p-473-12-x-2-cold-finished-1018.aspx
2. Block of aluminum (adjacent cell) - https://speedymetals.com/p-473-12-x-2-cold-finished-1018.aspx

Category	Item	Qty	Est. cost	Supplier / link	Why we need it
Barrier	Phlogopite mica, 0.51 mm	2 sheets	**$49.08**	ElectronTubeStore	Thin electrical + thermal barrier benchmark

Barrier	Ceramic-fiber paper, 1/8 in	2 sheets	**$20.00**	Discount Glass Supply	Cheap high-temperature insulation control
Barrier	304 stainless shim, 0.010 in	1	**$28.30**	Online Metals	Warwick-style metal barrier / face sheet
Barrier	3-mm silica-aerogel blanket	1	**~$70.00**	Aerogel-it	Very-low-conductivity benchmark; listed from €49 before shipping
Barrier	Intumescent/silicone fiberglass sample	1	**~$35.00**	Mid-Mountain ARMATEX SIF	Heat-triggered expanding barrier; combine with stainless
PCM	Sodium acetate trihydrate, 500 g	1	**$21.95**	The Science Company	MDPI-inspired phase-change experiment
PCM	Paraffin wax, 500 g	1	**$15.38**	Fisher Scientific	Second PCM component
Dummy cells	6061 aluminum, 3/4-in round bar × 12 in	2	**$15.44**	Online Metals	Machine cylindrical battery surrogates
Heating	12-V, 40-W cartridge heaters	4	**$22.00**	SeeMeCNC	Controlled heat inside dummy cells
Heating	12-V, 50-W silicone heating pad	1	**$23.64**	PMD Way	Alternative broad-area heat source
Power	12-V, 10-A Mean Well supply	1	**$27.70**	Bravo Electro	Heater/fan power if a lab supply isn't available
Control	PWM MOSFET heater modules	2	**$3.90**	Abra Electronics	Arduino heater control + spare

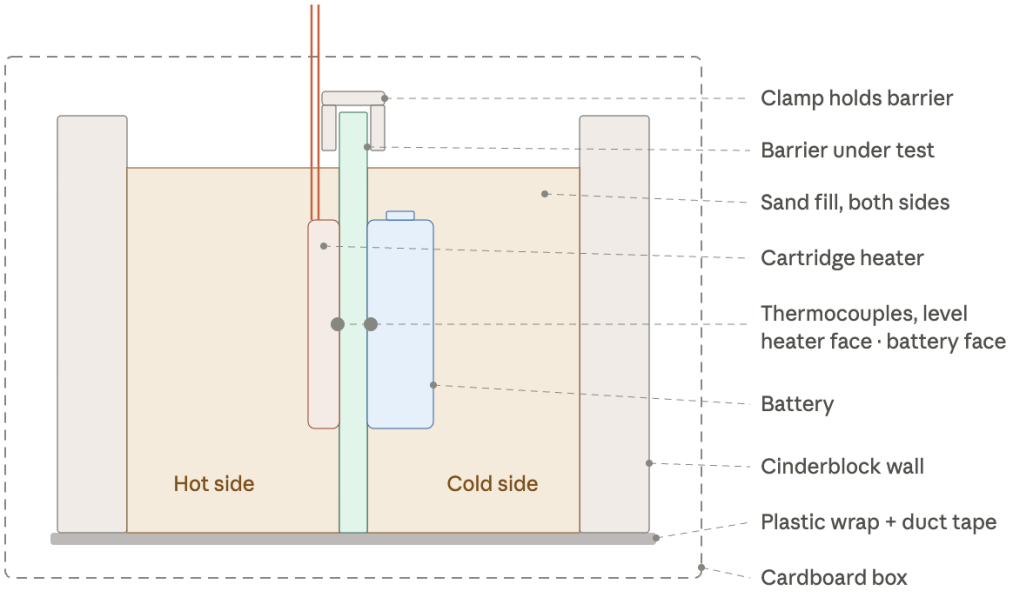
Venting	12-V blower fans	2-pack	**$8.19**	Blower-fan option	Forced-air / directed-vent comparison
DAQ	Arduino UNO R4 Minima	1	**$20.00**	Arduino	Data logging and heater/fan control
DAQ	MAX31850K thermocouple amplifiers	4	**$59.80**	Adafruit	Four simultaneous high-temperature channels
Sensors	Fiberglass K-type thermocouples	4	**$39.80**	Adafruit	Source, barrier-hot, barrier-cold and receiver temperatures
Wiring	Inline fuse holders + fuses	2	**$9.98**	Progressive Automations	Heater circuit protection
Wiring	Ceramic high-temp terminal block	1	**$7.00**	Duralite	Connections near heated fixture
Wiring	High-temp fiberglass wire sleeve	5 ft	**$10.00**	JDCustoms USA	Protect wiring near heaters
Battery hardware	18650 holder clips	8	**$7.60**	SparkFun	Removable connections for normal-cell tests
Fixture	Fasteners, spacers, wire, paste, misc.	allowance	**$15.00**	Lab/local stock	Maintain repeatable geometry/contact pressure
Batteries	Samsung 25R 18650	4	**$19.96**	18650 Battery Store	**Normal-operation validation only**, not abuse testing
Safety	Steel secondary guard	1	**$13.99**	Harbor Freight steel can	Guard around the heated surrogate fixture; do not seal around

					live-cell abuse tests
Software	Arduino IDE	—	**$0**	Arduino	Logging/control
Software	Python/Jupyter	—	**$0**	Python	Data reduction, plots, uncertainty/statistics
Software	Zotero	—	**$0**	Zotero	Literature management
Software	CAD/Onshape Education	—	**$0**	Onshape/institution software	Fixture and duct design
Research	Papers/library/standards	—	**$0 planned**	University/library access	Use institutional access rather than buying individual papers/standards
	Selected purchase subtotal		**$543.71**		
	8% contingency		**$43.50**		Shipping, tax, vendor quote variation
	Projected project spend		**$587.21**		**$62.79 remains**

Estimated Material Prices & Purchasing Links

- **1. Pre-Made CFRP Flat Plates / Sheets ($40 – $120 per sheet)**
  - *Application:* Used for rigid module end-plates, exterior structural housing walls, or inter-cell flat bulkheads.
  - *Example / Source:* DragonPlate Carbon Fiber Sheets offers standard (1 mm 12” x 12”) structural laminate plates for approximately **$107**. Thinner or smaller cut-offs can be found on industrial suppliers like McMaster-Carr for 40–80.
- **2. Carbon Fiber Sleeves / Tubing for Cylindrical Cells ($25 – $60)**
  - *Application:* Used to slide over individual cylindrical cells (e.g., 21700 or 18650 formats) to test custom lightweight thermal barrier sleeves.

    - *Example / Source:* CompositesCanada Carbon Fiber Braided Sleeves or similar specialty suppliers sell expandable carbon fiber sleeves by the yard for around **$25 to $50**.
- **3. DIY Vacuum Bagging / Resin Starter Kit ($80 – $150)**
    - *Application:* If you want to mold your own custom composite panels infused with fire-retardant epoxy or intumescent additives.
    - *Example / Source:* Fiber Glast Epoxy Resin & Carbon Fiber Starter Kits provide woven fabric, high-temp epoxy, and vacuum film components ranging from **$90 to $160**.



Hardware details

Fixture. The test sits in a cinderblock cavity, and the cinderblock sits in a cardboard box. The bottom of the cinderblock is sealed with plastic wrap and duct tape to keep the sand in. The barrier is clamped vertically in the middle of the cavity, dividing it into a hot side and a cold side. Both sides are filled with sand. Make the barrier span the full width of the cavity, or at least extend well past the heater and battery on every edge. Otherwise heat can travel through the sand around the barrier instead of through it.

Sand. Use dry sand. Damp sand absorbs heat as its water evaporates, which would make every barrier look better than it is. Use the same mass of sand on each side every run, and pack it the same way each time. Sand conducts better when packed tightly, so inconsistent packing adds scatter to your results. Let the sand cool fully between runs, or swap in fresh room-temperature sand. Sand holds a lot of heat.

Hot side. A high-temperature cartridge heater, rated for at least 700 °C, is pressed directly against the barrier's hot face. Power it through a variac. Since the variac is manual, you set the temperature by adjusting voltage while watching the heater thermocouple. Note your variac setting for each run. Ramp to about 600 °C and hold for 30 minutes. A real runaway spikes within seconds and then decays, so holding at peak is conservative. Report your ramp rate and the manual control as limitations of the method.

A bare cartridge heater only touches the barrier along a narrow strip, so it heats a line rather than a whole face the way a failing cell would. Note this in your report. Cartridge heaters also rely on

their surroundings to carry heat away, and sand does this poorly. Bring the voltage up gradually so the heater element doesn't overheat and burn out.

Cold side. The battery is pressed directly against the barrier's cold face. Use a fully discharged cell or an inert dummy cell, not a charged one (see Safety). The battery thermocouple sits on the face touching the barrier, which is the cell's hottest point. That makes the reading conservative, which is the right direction for this test.

Spacing. Because the heater and battery press directly on the barrier, the gap between them equals the barrier's thickness. Thinner barriers therefore put the heat source closer to the cell. That's realistic, since a thinner barrier in a module saves space, but keep it in mind when comparing materials of different thicknesses. Intumescents will expand against the sand rather than into open space, which is more confining than a real module. Note this as well.

Thermocouples. There are two: one on the heater and one on the battery, each on the side facing the barrier. Mount them at the same height, directly across the barrier from each other, so both measure the same heat path. Use a mineral-insulated K-type probe for the heater thermocouple, because the fiberglass-insulated kind fails above about 480 °C. Glass-insulated wire is fine for the battery thermocouple, held on with Kapton tape. Read both through MAX31856 boards at 1 Hz on the ESP32, which now only logs data and doesn't control the heater. Your detection subgroup can still reuse this data.

Test matrix

Cooldown is still the bottleneck. The sand and cinderblock hold more heat than the old fixture, so expect even longer waits. The intumescent and endothermic samples are destroyed in each test, so run the matrix in phases.

Phase	When	What	Runs
A: Screening	Weeks 2–3	Every barrier once, plus a no-barrier control and mica as the reference material	About 8
B: Finalists	Weeks 4–5	Top 3–4 barriers by figure of merit, 3 repeats each	About 12

C: Shock	Weeks 5–6	Finalist samples shocked at 20 g / 11 ms, then thermally tested and compared with unshocked results	About 6

Without altitude runs, Phase B is roughly half as long as before, which frees up time for extra repeats if you want tighter error bars. Cinderblocks, sand, and boxes are cheap, so a second fixture is easy to build. One can cool while the other runs.

For the no-barrier control, clamp nothing in the middle and use a thin sheet of a non-conducting material, or leave a fixed air gap with spacers so the sand from each side doesn't merge. For your barrier set: the control, mica, aerogel, the intumescent paint on thin steel, the expandable graphite/silicone composite at two loadings, the aluminum hydroxide/silicone composite, and an industry sample if you get one. Make enough of each finalist for three thermal runs plus shock samples.

Procedure for each run

1. Weigh the barrier sample and measure its thickness at several points.
2. Check that the plastic wrap and tape on the bottom of the cinderblock are intact.
3. Clamp the barrier in the middle of the cavity.
4. Press the heater against the hot face and the battery against the cold face. Check that both thermocouples are seated and level with each other.
5. Fill each side with the same mass of dry sand, packed the same way as previous runs.
6. Log 2 minutes of baseline.
7. Bring the variac up to reach setpoint. Start the 30-minute clock when the heater thermocouple reaches it.
8. If the battery thermocouple reaches 150 °C, cut power and record the time. That's a failed barrier, and heating further adds risk without adding information.
9. At 30 minutes, cut heater power and keep logging until the battery temperature has peaked and begun to fall. The sand stores heat and keeps feeding it to the battery after the heater is off, so this may take longer than the 10 minutes you used before.
10. After cooldown, remove the sand and gently brush the barrier clean. Weigh it again. The mass loss tells you how much gas it released, which matters for your purge subgroup. Photograph it and measure any char thickness or cracking.

Analysis

Plot battery temperature against time for every barrier on one graph, with a line at 150 °C. That single figure carries most of your story. Then make a scatter plot of t_{150} or peak battery temperature against areal density, so the lightest effective barriers stand out in one corner.

Finally, translate each result to the pack level. Barrier mass per cell divided by an assumed cell mass gives a parasitic mass fraction, and that converts directly into a Wh/kg penalty.

Safety

Use a discharged or dummy battery. If a charged lithium cell gets hot enough, it can go into real thermal runaway and vent flammable, toxic gas inside a cardboard box. Even with a discharged cell, stop at 150 °C on the battery thermocouple.

Run tests in a fume hood or outdoors. Hot silicone, intumescents, and graphite release smoke and gases, sometimes including ammonia.

Keep the heater leads and anything hot away from the cardboard. Cardboard can ignite well below your 600 °C setpoint. The plastic wrap on the bottom will melt at roughly 100 °C, so check it after each run and replace it if it's softened. A tray under the box will catch sand if the seal fails.

Variac-powered cartridge heaters run on 120 V. Have someone qualified check the wiring, plug in through a GFCI, and keep a kill switch within reach. Ideally, add an independent over-temperature cutoff on the heater too. Keep an extinguisher and a bucket of dry sand nearby, and never leave a run unattended.

Thermocouple Check

Purpose: confirm that both thermocouples read correctly against a reference thermometer.

Method:

1. Fill a beaker with room-temperature water.
2. Place both thermocouples in the water and let the readings settle.
3. Measure the water temperature with a reference thermometer.

Results:

Sensor	Reading (°C)	Difference from thermometer (°C)
Thermometer (reference)	21.9	—
T1	21.8	−0.1
T2	22.2	+0.3

Conclusion: both thermocouples agree with the reference within 0.3°C, well inside the standard type K tolerance of ±2.2°C. No correction needed.

1. Preparation (once)

1. **Dry the sand** in an oven at 110–150°C for 1–2 hours and let it cool.
2. **Check the thermocouples** in room-temperature water against a reference thermometer (done: both within 0.3°C).
3. **Build the barrier boxes:** cut and fold the mica boxes, then wrap them with aerogel, joined with stainless tie wire.
    - Layer order, hot to cold: rod → mica (box 1) → intumescent coat, if used → aerogel → mica (box 2) → cell
4. **Prepare the rod:** fit the cartridge heater snugly in the reamed hole. Attach T2 to the rod surface facing the barrier, and T3 to cell 2's surface facing the barrier.
5. **Set up the electrical side:** grounded rod and heater sheath, GFCI outlet, fuse, variac taped at 120V maximum, multimeter on the variac output and an ammeter or clamp meter for current.

2. Burn-in and calibration (aluminum dummy cell)

1. **Assemble the setup:** rod and dummy in their boxes, sand in the corners, outer insulation wrap, boxes clamped together the same way every time.
2. **Run outdoors or under ventilation.** The first heating burns off the mica binder and any residue on the aerogel or sand.
3. **Boost:** run at 100V from room temperature and record the time for T2 to reach 480°C. That time is y.
4. **Find the hold voltage:** switch to 55V. Adjust 3–5V at a time, waiting about 10 minutes after each change, until T2 holds steady near 500°C. That voltage is Y.
5. **Verify:** cool fully, then run the full recipe from cold (X = 100V for y minutes, then Y for the rest). Confirm T2 lands at about 490–510°C.
6. **Record the "before" condition of the barrier:** photos, mass (without sand) and thickness at 3 marked points.

Technology Constraints

Technology Readiness Level: 4

- Airplane simulation: level 5

Application: all-electric, battery-powered passenger aircraft (commonly categorized as electric Vertical Take-Off and Landing [eVTOL], urban air mobility [UAM], or regional electric aircraft)

1. **Thermal Non-Propagation Thresholds: Single-Cell Thermal Runaway Test** (RTCA DO-311A Section 2.4.5 / FAA Special Conditions) - Thermal insulation must keep adjacent cells safely below **+150 C** (or below the cell manufacturer's designated thermal runaway onset temperature, typically **+150 C to +180 C)** at simulated pressure altitudes up to **70,000 ft** (DO-160G Section 4 & 5). There should be zero flame propagation and temperature maintenance for ≥ **30 minutes.**
2. **Post-Crash Survivability:** a 15.2 m drop and not pose a risk to human life, i.e. not leak gas, not fly into the cockpit, not become deformed and therefore a potential bomb. Needs to survive crash conditions 6g (operational shock) to 20 g (crash impulse) for 11 ms duration (DO-160G Section 7)
3. **High gravimetric energy density (primary constraint):** The battery pack must provide sufficient usable energy per unit mass to meet the aircraft's range and payload requirements. For a 19-seat, all-electric commuter aircraft, a representative target is approximately 650 Wh/kg at the pack level. Weight requirement on module weight.
4. **Early Thermal Runaway Detection and Fail-Safe Warning Systems:** Regulatory basis: FAA AC 20-184A (Appendix C, Special Conditions 7–9; Appendices D and G), 14 CFR §§25.1309 and 25.1322, RTCA DO-311A §2.1.4.2, RTCA DO-160G, and EUROCAE ED-289. The battery monitoring system must detect abnormal cell temperature, voltage, or gas emissions and **alert the pilot within 10 seconds**, with critical sensor **readings updated at least once per second (1 Hz)** and **automatic charging disconnection** upon confirmed overtemperature or battery failure. Detection and warning systems must undergo RTCA DO-160G environmental testing and fault-injection testing, demonstrating reliable operation under temperature extremes, vibration, altitude changes, and individual sensor failures, with zero missed required warnings across the qualification test matrix.

Budget: $650

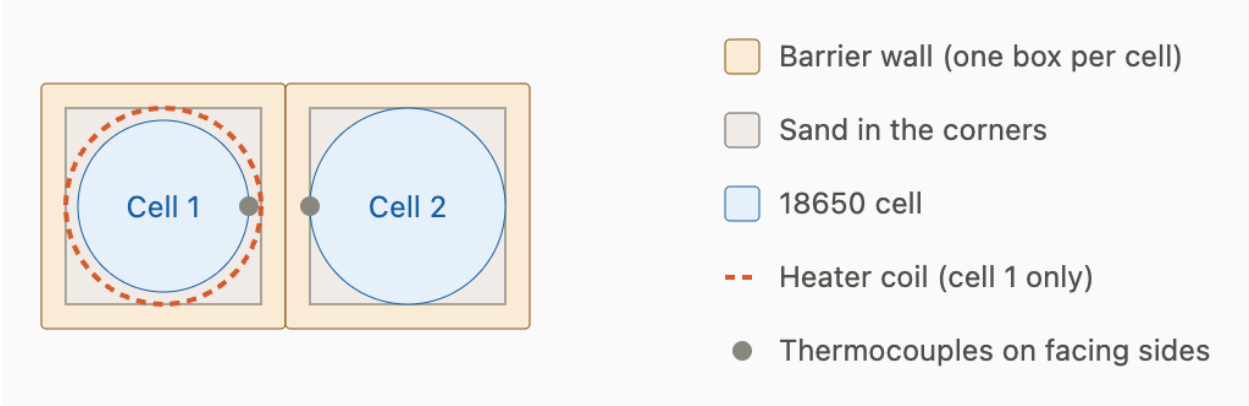
Material Densities

Here is the ranking of all the physical barrier and structural materials discussed, ordered from **lowest density (lightest)** to **highest density (heaviest)**:

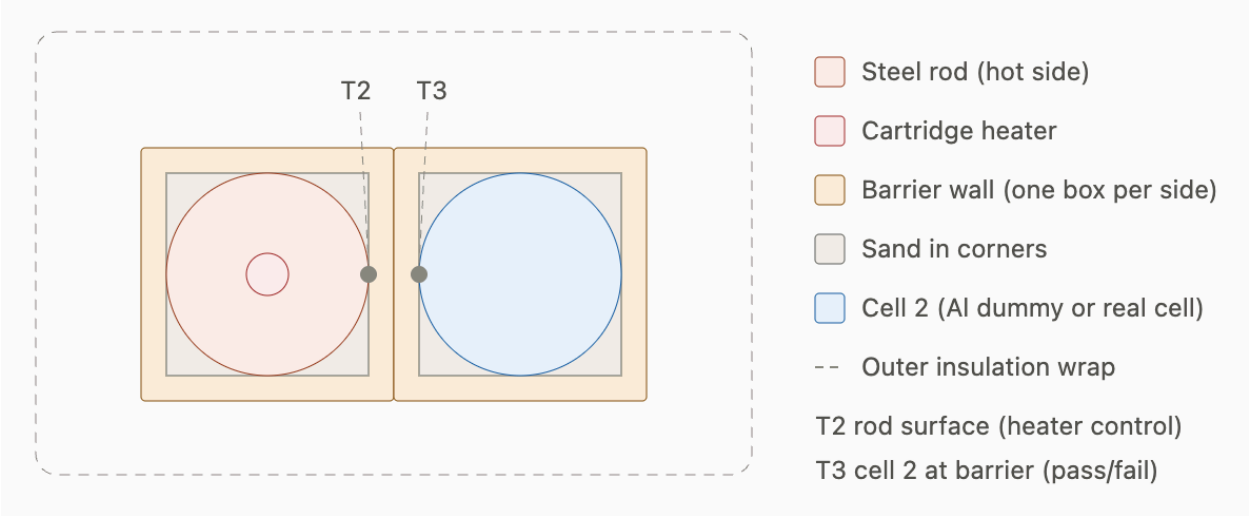
1. **Nanoporous Silica Aerogel Sheets**
    - **Density:** $\approx 0.003 \; to \; 0.15 g/cm^3$ (Bulk density)
    - **Note:** The lightest material available, owing to its extremely high porosity (approx 99%).
2. **Ceramic Fiber Paper / Mats (Aluminosilicate)**
    - **Density:** $\approx 0.20 \; to \; 0.50 g/cm^3$ (Bulk density)
    - **Note:** Highly porous fibrous insulation engineered for high-temperature resistance.
3. **Phase-Change Materials (PCMs, e.g., Paraffin Wax Matrices)**
    - **Density:** $\approx 0.80 \; to \; 0.90 g/cm^3$ (Solid state)
    - **Note:** Organic waxes or salt hydrates used for latent heat enthalpy buffering.
4. **Carbon Fiber Reinforced Polymer (CFRP) Composite Plates / Skins**
    - **Density:** $\approx 1.50 \; to \; 1.60 g/cm^3$
    - **Note:** High-strength, lightweight structural composite used for aerospace enclosures and housings.
5. **Intumescent Polymers / Coatings / Foams**
    - **Density:** $\approx 1.10 \; to \; 1.40 g/cm^3$
    - **Note:** Standard polymer binder and resin densities prior to volumetric expansion under heat.
6. **Aluminum Hydroxide ($Al(OH)_3$)**
    - **Density:** $\approx 2.42 g/cm^3$ (Material density)
    - **Note:** Used as a sacrificial endothermic filler compound that releases water vapor upon thermal decomposition.
7. **Basalt / "Lava Rock" Fiber Wraps**
    - **Density:** $\approx 2.67 g/cm^3$ (Solid mineral density; textile bulk density is lower)
    - **Note:** Volcanic rock-derived mineral fibers offering high radiant heat and flame resistance.
8. **Phlogopite & Muscovite Mica Shields**
    - **Density:** $\approx 2.70 \; to \; 2.85 g/cm^3$ (Crystal/sheet density)
    - **Note:** Dense mineral plates providing rigid dielectric separation and direct flame-jet defense.

Comparison of Experiment Setups - 10/16

2 batteries



Cartridge + Battery



Option A: heater coil wrapped around a real cell 1

Pros

- Realistic geometry and materials on both sides.
- Cells are cheap (the 25R is $4.99), and the setup is simple to build.

Cons

- **It can't get hot enough.** A charged cell starts heating itself at roughly 80–120 °C, so to stay safely out of runaway you'd have to stay below that. Your barrier would then be holding back less than 100 °C instead of a runaway-like 500 °C or more, which makes the test far too easy and not representative.

- **Discharging cell 1 doesn't fully fix it.** An empty cell won't run away violently, but well above 150 °C it will still vent flammable electrolyte vapor.
- **The coil sits between the cell and the wall.** It changes the contact you're trying to measure, and the coil runs hotter than the cell, so you can't be sure what temperature the barrier actually sees.

Option B: cartridge heater in place of cell 1

Pros

- It can safely reach about 500 °C and hold it for the full 30 minutes, which is what your requirement calls for.
- Temperature control by PID makes runs repeatable from one barrier to the next: or open control
- It's reusable, with no hazardous chemistry.

Cons

- **It needs a sleeve.** A 6 mm cartridge heater doesn't fill an 18.4 mm box. Machine a steel rod about 18 mm in diameter and 65 mm long, with a 6 mm hole down the center for the heater, so the "hot cell" touches the walls the same way a real cell would.
- **Not a real cell.** There's no internal heat generation, venting, or flame, the same limitation as the block design. Note it in your report.

Final test with cartridge heater
Detecting change in temperature for neighboring
Barrier: extending time

Definition of thermal runaway for cartridge heater and/or adjacent cell

- How fast of temperature increase
- How long of temperature
- Ex: In the course of the “lab period/number of time”, the temperature doesn’t reach _

Acceptance Criteria

Test setup (apply to every run)

- **Cell 1 (heat source):** a steel rod heated by a cartridge heater, standing in for a cell in thermal runaway. It is held at T2 = 500°C ±10°C, measured on the rod surface at the barrier.
- **Cell 2:** a Samsung [model] lithium-ion cell at [100]% state of charge. Use a fresh cell for every run.
- **Hold time:** the 30-minute clock starts when T2 first reaches 500°C.
- **Kept the same every run:** barrier build, sand fill, clamping and spacers, thermocouple attachment, outer insulation wrap, and starting temperature (room temperature ±[3]°C).
- **Invalid runs:** a run doesn’t count if T2 leaves the 490–510°C range after reaching setpoint, or if a thermocouple comes loose. Treat either as a setup problem, not a barrier failure.

1. The barrier prevents thermal runaway from spreading to cell 2 for 30 minutes

- **Definition:** thermal runaway is when a cell’s internal chemical reactions produce heat faster than it can escape. The cell then keeps heating itself, ending in venting, fire or rupture. In this test, the barrier must keep cell 2 below the temperature where its own self-heating would begin.
- **Pass, all of the following for the full 30-minute hold:**
    - T3 (cell 2 surface at the barrier) stays below [100°C].
        - Why 100°C: in lithium-ion cells, the protective layer on the anode starts breaking down at about 80–120°C, separators begin melting at about 130°C, and self-heating usually starts somewhere between 90 and 150°C. A 100°C limit leaves margin. **Confirm it against the Samsung datasheet**.
    - Cell 2 shows no venting, smoke, swelling or fire.
- **On failure:** record the time at which T3 crossed the limit, or when venting started.

2. The barrier is still usable after one run

- **Reusable vs. consumable parts:**
    - The mica box is expected to be reusable.
    - [The aerogel wrap is replaced each run, or is expected to be reusable; decide which.]
- **Burn-in first:** do the "before" measurements after one burn-in run, because the first heating burns off the mica's binder.
- **Pass, for the reusable parts:**
    - No through-cracks, holes or peeling layers on the barrier face, documented with before-and-after photos at the same angle and lighting.
    - The box holds its shape and can be taken apart and reassembled without breaking.

3. The setup and results are reproducible

- **Number of runs:** at least 3 valid runs with the same configuration, each with a fresh Samsung cell.
- **Main metric:** the rise in T3 over the hold, $\Delta T3$ = T3 at 30 min − T3 at start. Each run's $\Delta T3$ must be within ±10% of the average of all runs.
    - Use the temperature rise rather than the raw T3 reading, because a percentage of a raw °C value depends on the starting temperature.
- **Secondary metric:** the time for T3 to reach [50°C] is within ±10% across runs.

Safety requirements (before any run with a real cell)

- **Location:** outdoors or in a fume hood or fire-rated enclosure. A cell in a runaway releases flammable, toxic gases, including hydrogen fluoride.
- **Manual monitoring:** T2, T3 and cell 1's voltage are logged remotely, and operators will open-control
- **Fire response:** plenty of water ready nearby for cooling the cell.

Limitations (state in the report)

- **Heat only:** the heater reproduces conducted heat only. It does not vent gas, eject material or produce flame jets like a real cell in a runaway, so the test doesn't show how the barrier handles those.
- **Lower peak, longer hold:** 500°C is below the 600–800°C+ peak a real Samsung cell can reach. But holding it for 30 minutes delivers far more total heat than a real event, where the peak lasts only minutes. Overall, the test is conservative for heat soaking through the barrier.